Imparting Full-Duplex Wireless Cellular Communication in 5G Network Using Apache Spark Engine

Zahraa A. Jaaz
College of Computing and Informatics, Universiti Tenaga Nasional (UNITEN), Malaysia. zahraa.jaaz@nahrainuniv.edu.iq
College of Science - Computer department - AlNahrain University, Baghdad-Iraq.
https://orcid.org/0000-0003-0928-679X

Inteasar Yaseen Khudhair
Department of computer Science , Basic Education College, University of Diyala. Inteasar_yassin@yahoo.com

Israa Al_Barazanchi
College of Computing and Informatics, Universiti Tenaga Nasional (UNITEN), Malaysia. Israa44444@gmail.com
Computer Techniques Engineering Department, Baghdad College of Economic Sciences University, Baghdad - Iraq
https://orcid.org/0000-0002-6798-6295

Hala Shaker Mehdy
College of Education - Computer Science , Al-Mustansiriya University - Baghdad, Iraq.
hmprogeram@yahoo.com
College of Computing and Informatics, Universiti Tenaga Nasional (UNITEN), Malaysia.

***Abstract*— With regard to the forthcoming requirements for mobile services in 5G networks, several new technologies have recently become the focus of leading-edge research. One of those is in-band full-duplex communications in processing data with bigdata based Apache spark. The idea is to simply employ the same frequency band to simultaneously transmit and receive information, allowing more spectrally efficient communications when compared to the traditional half-duplex or out-of-band full-duplex counterparts. By breaking a long-held assumption in wireless communications, in-band full-duplex. this paper aims to study wireless communication in 5G network with use of bigdata based Apache spark. Particularly, apache spark suppression filters are studied, as well as feedback adaptive filtering is proposed for relay systems with multiple-input multiple-out (MIMO) antennas. In this case, the relay system energy efficiency is maximized by finding the optimal transmit powers, while maintaining a certain individual link quality. For this scenario, the effect of massive MIMO is likewise addressed, and an algorithm that maximizes the system total achievable rate is derived. 5G technologies have pressing security challenges to secure data integrity and privacy in critical wireless communications requiring extensive research before implementing these technologies into use. This paper examines the new possibilities 5G networks offer for wireless network communication with high bandwidth of 90 GHz. The prediction of 5G network bandwidth maximum recoded at 90 GHz on 5 nodes of apache spark at 80% of data trained with total 12 nodes. The main question is will the 5G technology offers sufficient performance for wireless communication.**

Keywords—5G, in-band full-duplex, MIMO, big data, wireless communications, Apache spark, spectrum allocation

I. Introduction

The constant demand for higher rates and lower latencies has already driven the scientific community and industry to pursue a new generation of mobile communications. Simply known as the fifth generation (5G), and with expected release date in 2021, telecommunications are about to go through a deep change in their foundations. One of those central changes is the long held, but now obsolete, assumption that radios can only simultaneously transmit and receive in different frequency bands on the same channel as mentioned in [1]. Thus, the concept of in-band full-duplex communication, i.e., simultaneously employing the same frequency band to transmit and receive data in wireless nodes, will certainly incorporate this new generation, providing a leap forward in terms of spectral usage efficiency. Moreover, when integrated under a physical layer network perspective or when combined with the rise in the number of antennas at base stations, its gains are beyond those expected few years ago. Furthermore, the orthogonal properties of large-scale arrays are resorted to, such that an extra level of mitigation is achieved.

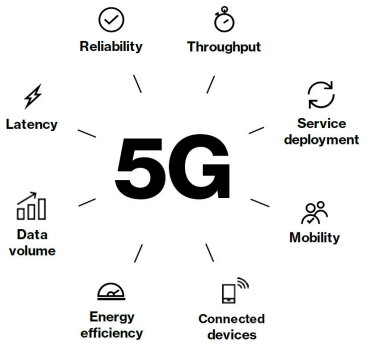


Fig. 1. Some of the most common use-cases of 5th generation wireless network [2].

In recent decades, the topic of mobile communication has been a subject of interest to the research and industry communities. Several different technologies were studied so that a standard system could be implemented, allowing wireless communications to be a reality today. In the last 20 years, different major market requirements have appeared, pushing the telecommunication industry to develop new solutions. Firstly, the need for wireless real-time voice communication boosted the first generation (1G) of analog cellular transmissions as mentioned in [3]. Then, short message service (SMS) was incorporated in a digital system

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

that led to the second mobile generation (2G), the first to use a digital transmission. Due to the popularity of laptops and wireless local access networks (WLAN), wireless data connectivity became a hot topic and with it the third generation (3G) emerged as mentioned in [4]. Currently, the fourth generation (4G) of wireless connectivity has been derived from the need for faster links, aiming to provide higher bandwidth access anywhere

A. Problem Statement

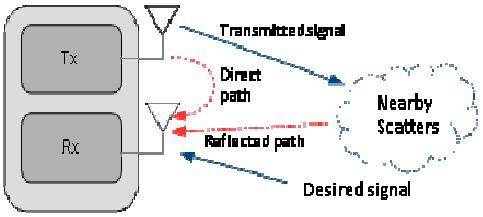
In this paper, in-band full-duplex exchange of information under a MIMO and under a physical wireless network perspective is assessed with big data. The main objective is to study and propose some techniques for in-band full-duplex systems for 5G network using the big data techniques, that take tools from current state-of-the-art technologies described in this work. Self-interference mitigation methods in big data, relaying techniques, physical layer networking and the use of massive antennas are further studied and proposed to integrate in-band full-duplex systems in 5G networks. This work focuses mainly on the theoretical side of telecommunications, always assuming transmissions in the digital baseband domain in big data. A simulation framework is then developed to evaluate and confirm the performance of the proposed 5G system. A research combining the in-band full-duplex transmission in a relay station with the use of a massive number of antennas of 5G network. A communication link is established between several pairs through the relay system in big data. This relay incorporates self-interference suppression filters, as well as detector and pre-coder filters, whose performance is evaluated at the relay and at the destinations with BER curves. Further, an optimization problem is solved that ensures a minimum quality link for the pairs, while minimizing the total energy spent.

B. Aim of Contribution

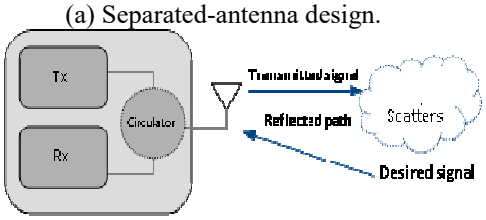
With the widespread use of 5G technology, the number of users on communication network have also increased as given in [5]. According to Gartner Research in [6], the number of users on communication devices by 2050 will hit 18 billion. The scope of this paper is beyond the operations of 5th generation networks as we determined its usage and criteria under the severe weather conditions. This study has prominence because it is state-of-the-art work performed under the research parameters of 5th generation networks using big data technology. Thus, we proposed the big data based on the dataset acquired. Furthermore, some variations in the context of traffic profile are performed, in terms of service throughput, service share percentages and different traffic profile configuration shares in some municipalities as mentioned in [7]. In terms of licensing and legalizing the use of big data frequency for possible because of its designing and development with the open-source material and methodology with high transmission power as well as high transmission rate under uncertain weather conditions using MIMO input/output antennas used with Down-Link (DL). Firstly, this work will fill the gap of research where a 5G technology, which is under field testing and for a part never been tested under different weather conditions using both of big data toolbox and signal processing toolbox. This research paper provides an investigation of 5G technology dimensioning as objective for different weather conditions, while also permitting changes in input parameters for different network deployment configurations from Cumulative Distribution Function (CDF). The most advanced big data techniques and model for 5th generation communication network as well as for coverage and capacity are defined in this work. Besides these essential elements for cellular network planning there is the usage of 5G network in terms of density/volume and share/mix profile throughout a typical load day are also been discussed.

II. Background

The first approach to mitigate self-interference was based solely on propagation properties of electro-magnetic waves in 5G networks. The need to connect terminals that are too far to establish reliable communication brought about the idea of using relay stations as mentioned in [8]. In order to increase the coverage of networks, relays can operate within the same bandwidth and act as passive amplifiers by receiving and retransmitting an amplified version of the desired signal, known in literature as amplify-and-forward (AF). Therefore, wireless propagation insulation techniques were developed, aiming for separation of the receiving side from the transmitting side of the terminal as mentioned in [9].



(a) Separated-antenna design.



(b) Shared-antenna design.

Fig. 2. Different possibilities to mitigate self-interference in 5G networks using wave propagation properties [9].

The first, separated-antenna design, consists of using different antennas to transmit and receive, as shown in Fig. 2(a). Using this procedure, it is then possible to suppress interference by path loss attenuation, increasing the antennas distance to further increase the attenuation or by placing absorptive material between them as given in [10]. However, this approach is limited by the terminal design that today plays a serious and important part in commercial products. Cross-polarization solutions are also interesting possibilities to insulate antennas and enhance the interference suppression as given in [11], for example, by transmitting in vertical polarization and by receiving in horizontal polarization. Another possibility is to exploit the antenna radiation pattern, carefully placing the receive antennas in radiation null points of the transmit antenna array as given in [12]. The second one, shared-antenna design, in Fig. 1(b), consists of using the same antenna to transmit and receive. This is possible by means of a circulator, a circuit that routes the signals from the transmit chain to the antenna and from the antenna to the receive chain as given in [13]. In the case of a relay station, it is common to place antennas back-to-back, i.e., arraying

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

antennas together that connect each relay end. Furthermore, one can apply techniques such as wave traps as given in [14], which extend the relay isolation by connecting short circuit resonant transmission lines, such as band-gap structures as mentioned in [15]. These materials have high surface impedance and for that reason attenuate the wave propagation. Also, one may use slots on the ground plane [16], which consist of internal dual band diversity antennas with dual frequency notched resonators to enhance electromagnetic attenuation. In regard to carrier aggregation techniques, they are to provide the operator with the opportunity of combining and operating separate, often non-contiguous, blocks of spectrum as one block with the goal of maximizing available and future spectrum, in order to provide a higher level of performance consistency as mentioned in [17]. There are three configurations for carrier aggregation:

• Contiguous frequency band: carriers are contiguous and located within the same frequency band. In this case the MT can handle signals using a single trans-receiver.

• Non-contiguous frequency band | intra-band: carriers are located within the same frequency bands, but they are not adjacent to each other. In this case the MT may require a separate transceiver for each carrier.

• Non-contiguous frequency band | inter-band: carriers are in different, distinct parts of the radio spectrum. It requires a transceiver for each carrier and the MT must operate in multiple bands at the same time. It is very useful for operators with limited or fragmented spectrum, though at the cost of increased system design complexity, cost and power consumption.

Due to a commercial competition for spectrum at lower bands, operators have often to use carrier aggregation techniques to make use of their non-contiguous frequency band. Although there have been successful tests in LTE with advanced carrier aggregation producing data rates up to 600 Mbit/s per user in controlled environments, they were based on higher MIMO order (8x8) and on limited/research spectrum, thus it does not and cannot still apply to most of real-life deployments as mentioned in [18]. Regarding bandwidth in 5G, the main difference with LTE (4G) is that 5G will consider a single bandwidth of up to 800 MHz, whereas in LTE the maximum is 10 GHz, for single channels. Hence, this is a clear indicator of the potential of 5G to achieve even higher data rates since with carrier aggregation [19], in the future, the maximum achievable bandwidth will be even considerably higher than those in LTE with the same techniques.

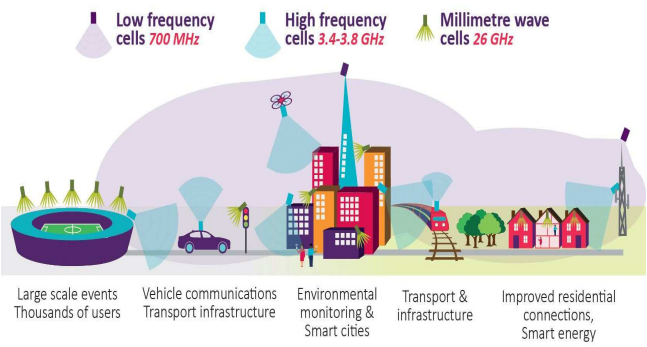


Fig. 3. The 5th Generation network improving the modern society through wireless communication [19].

As previously discussed, the model can take any numerology configuration although only the one with highest distance is considered for coverage distance purpose as mentioned in [20]. If numerologies 0, 1 and 2 are selected, coverage estimation will always be based on numerology and consequently the same applies for computing the maximum achievable throughput at the cell edge. Although there is a liberty to pick any numerology configuration, there are certain restrictions imposed by 5G. Essentially there are numerologies that are not supported by some pairs of frequency/bandwidth.

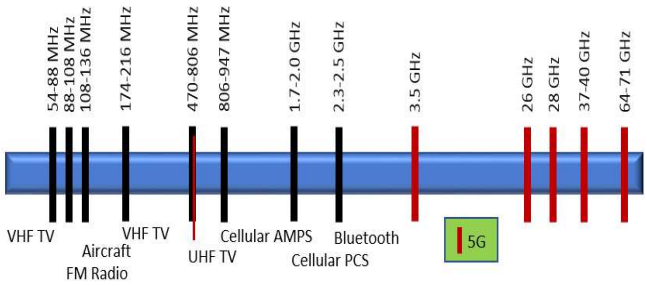


Fig. 4. The radio spectrum in 5th generation wireless networks [20].

In this sense, considering the user population randomly distributed in the cell and the capacity possibilities in terms of throughput from weather regions and numerologies, the model picks the most efficient 5G in use ratio for every service.

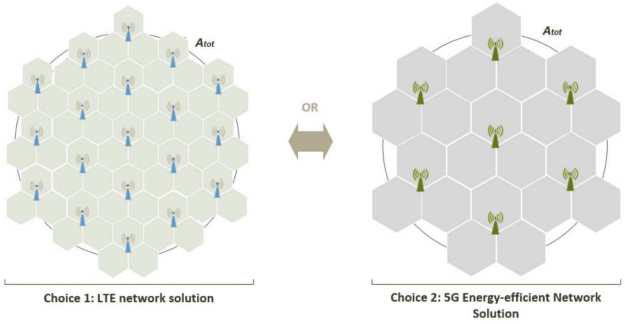


Fig. 5. The difference between the LTE network and 5G network in terms of network solution [20].

III. Methodology

Current big data with 5G-connectivity have been able to transmit selected data, such as location, speed, and turning angle to the network to be used for automated driving. However, big data are producing an enormous amount of data through the multiple sensors and cameras that the 5G-network throughput is not able to cope with. On top of the sensor data, big data are using data for various applications such as infotainment and safety functions. Due to the nature of applications' data usage, the requirements for each connection are different. Infotainment such as streaming multiple video inputs simultaneously, requires high data throughput without the need for low latency or high reliability. Safety functions require high availability, ultra-low latency, and high throughput if uncompressed data streams are used.

In broader terms, regarding general 5G network schemes is focused on the study of mill-metric waves in the access radio, front haul and backhaul contexts, in a variety of scenarios. The group has performed analysis on the use of

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

mm-waves in micro-cells for dense radio-access areas in indoors and outdoors, to infer the technical feasibility of network increase capacity. The fantastic 5G project has researched the primary key indicators needed to be achieved to improve capacity, coverage, reliability, and latency for networks operating below 3.5 GHz, based on LTE and 5G for model simulations. We developed a model that computes capacity and coverage for mill-metric waves in the 10 GHz band, in an urban scenario. The implementation of mm-waves for future 5G networks, having as underlying modelling interface an opensource MATLAB software, capable of simulating 5G-network parameters with a 100 GHz bandwidth. At last, the works that relate more to the frequency spectrum in the scope of this research paper and network characteristics, such as path-loss or MIMO, presents a simplified but comprehensive overview on 5G deployment considerations, from use-cases to frequency bands' implementation roadway, as well as interoperability architectures with LTE core network.

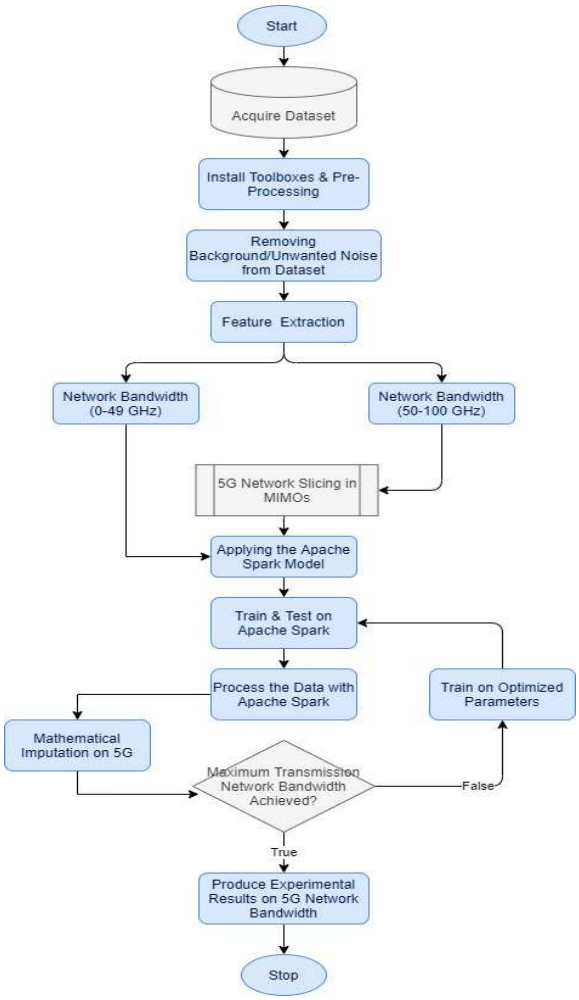


Fig. 6. Flow diagram of approach being followed.

The dataset has been acquired from an open-source repository crawdad: http://crawdad.org/keyword-5G.html. A dataset is composed of rows and columns, where, usually, each row represents an observation and each column a different variable. The 5G wireless transmissions have provided the tools to design faster and more reliable systems for wireless communications networks. The idea is to exploit the space diversity introduced by using multiple antennas, in order to increase user capacity. Using a 5G MIMO channel in this dataset for communication provides several data streams sent simultaneously, called spatial multiplexing, where information is processed and stored by apache spark to serve different users. The key concept is to take advantage of the multiple scattered uncorrelated signals that arrive via different paths at the receiver to extract the information under transmission. The 5G mobile communication technology that is about to be harnessed into use during 2021 has been an anticipated evolution of the currently used 4G technology. The extensive performance and multiple new services are enabling innovations in multiple different verticals. The 5G wireless mobile communication with apache spark is one of the verticals that benefits from this technological leap. The research evaluates if the 5G technology offers sufficient performance for the fully automated communication using the big data based technology for processing. The big data based apache spark will serve the end-users with ultra-low latency and highly reliable communication that is required by autonomous devices to communicate with smart city infrastructure. The apache spark offers high data throughput that can be used to transfer data-intensive sensor input to the cloud for further analysis.

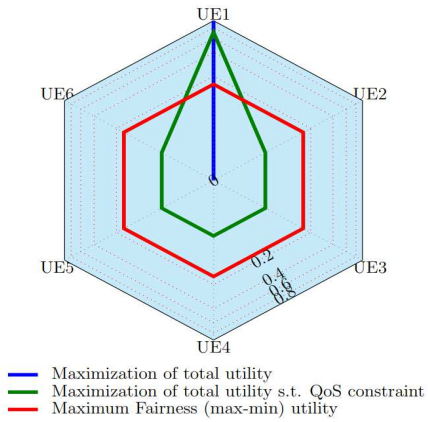


Fig. 7. The maximum total utility of UE in 5G network with QoS.

The improvements in the performance and changes in the services and require changes in the network architecture compared to the 4G technology. These changes, however, are adding challenges to the network design, especially to the security of the network. The pressure to deliver high performance might lead to compromises for the security and privacy of the network. For example, extensive authentication adds latency and limits the throughput of the communication. The architectural changes might also open up unexpected attack surfaces in services such as network slicing. The research focus was on the communication features of the proposed 5G protocol. The proposed communication and parallel processing protocol of big data for 5G was apache spark found to be prone to replay communication. The first tool applied in this step is a network traffic flow crawler, a tool which gather information, content and the whole structure of a determined set of network traffic flow pages. We are going to talk briefly about the whole process later in this chapter and also deeply in a more specific section. Besides extracting the features from a dataset, which is going to be well explained in this work, we have also written some code to extract information from those files, the features to analyses. To do so, we have used the Apache Spark environment in big data, a very concise and powerful framework for wireless sensor

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

network. However, we only used a small portion of data in the beginning, but increased the amount over time.

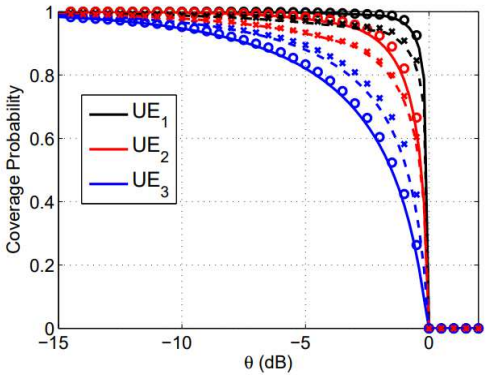


Fig. 8. The UE of 5G network with coverage probability of signals higher than 50GHz for communication case rate.

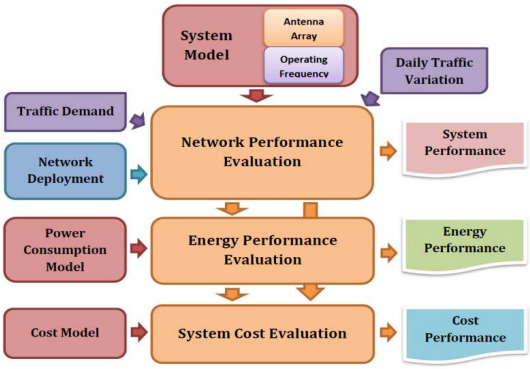


Fig. 9. The system model and performance evaluation model for network traffic and power consumption for 5G.

IV. Results

A new kind of big databased Apache spark is introduced with optimized version of it for the purpose of investigating the different wireless communication effects on 5th generation network with a greater number of training epochs. This big databased Apache spark is used in this work have distinguished size and number of nodes and its formula for the threshold has been modified – a new parameter for different wireless communication is added to it and an optimization is made for this new parameter on MATLAB (Using Parallel Computing Toolbox and Signal Processing Toolbox). Some details, such as activation function choice and apache spark optimization algorithm choice are this work's original contributions for the achievement of goal. We performed the feature extraction and selection process using apache spark is less time just because of optimization before splitting the dataset for training and testing. By separating 80 percent of the dataset for training and 20 percent of the dataset is parted as testing set. As in past there is no specific application been developed for the purpose of investigating the wireless communication using the apache spark as well as investigating the wireless communication effects on 5th generation networks with CDF. The application which is designed and developed solely for fulfilling the identifying the communication conditions and investigating the performance of 5th generation networks communication for the operational purpose using the big data. This application demonstrates that 5th generation network being firstly used for wireless communication based on the dataset. As it would be expected, within the same environment type the maximum cell radius is higher for the 50 GHz band and lower for the 100 GHz band, and as the numerology configuration increases the cell radius is reduced using apache spark.

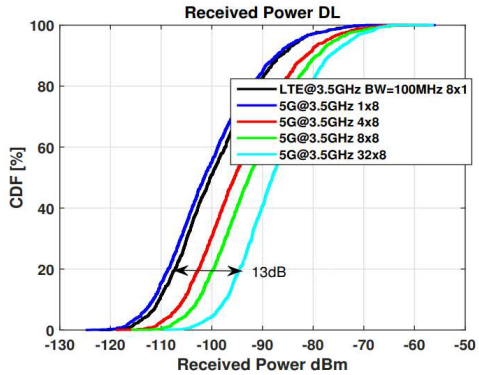


Fig. 10. The received power DL of 5G and LTE with CDF for wireless communication in dBm.

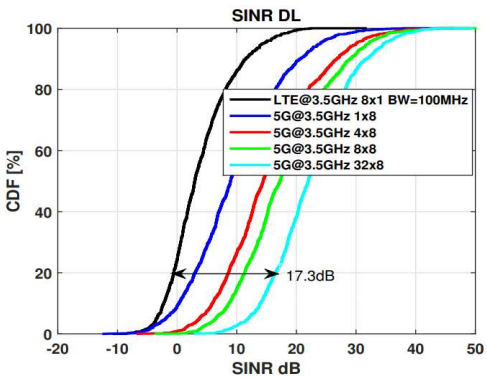


Fig. 11. The received signal SINR DL of 5G and LTE with CDF for wireless communication in dB.

Furthermore, this application shows that the accuracy of apache spark on 5th generation network, which can have mitigated to some degree of optimization due to a greater number of nodes in the apache spark and achieving a high accuracy for wireless communication. We have used an effective and efficient application of MATLAB R2019a because it provides parallel processing toolbox, machine learning toolbox as well as signal processing toolbox for the implementation of this research work.

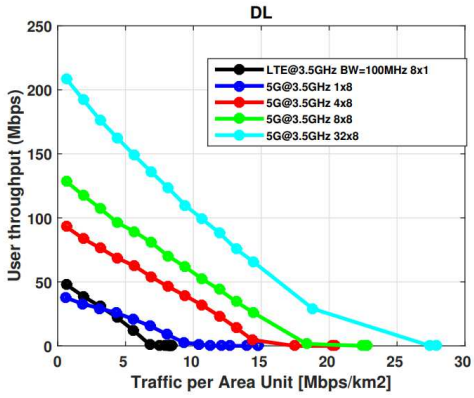


Fig. 12. The user throughput in comparison with traffic per area unit in kilometer square for DL in 5G and LTE for wireless communication.

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

Nevertheless, most systems that operate in broadband need to deal with frequency-selective DL channels, since the previous assumption is no longer valid. Modern wireless applications require faster data rates where the flat fading assumption cannot be assumed, enhancing the study of techniques to overcome this problem with lower CDF in 5G LTE network.

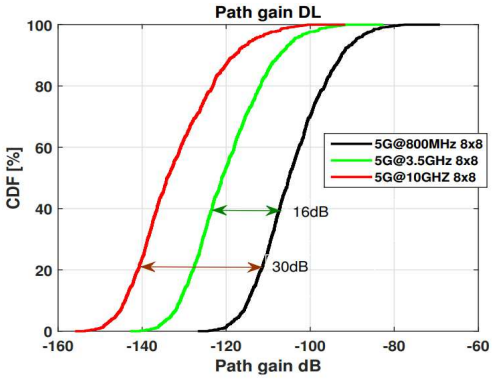


Fig. 13. The path gains DL for 5G network range from 800MHz to 10GHz with percentile CDF function for wireless communication.

The frequency-selective channels may actually achieve higher rates than flat channels. This issue has been analyzed in this work, where the big data based apache spark develop a simulation-based method to compute the achievable rate of CDF systems operating under different communication conditions.

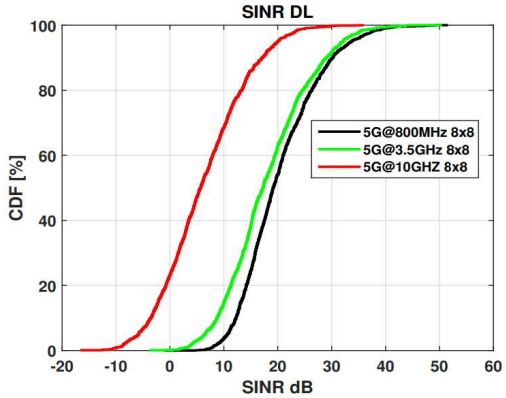


Fig. 14. The received signal SINR DL for 5G network range from 800MHz to 10GHz with percentile CDF for wireless communication in dB.

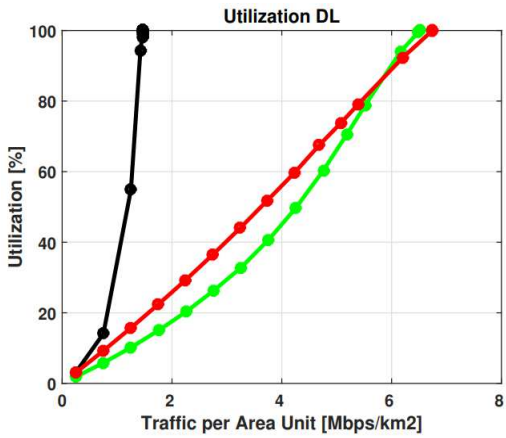


Fig. 15. The utilization DL with traffic per area unit in kilometer square for wireless communication.

V. Discussion

The wireless 5G networks is driving the search for disruptive technologies capable of corresponding with these new demands with big data based apache spark. Among them and with great potential, lies in-band full-duplex communications, which will double the spectral efficiency of current transmissions and will facilitate management of the available spectrum. However, self-inflicted interference poses a great challenge to the deployment of this technology. If not properly dealt with, this so-called self-interference may create unbearable signal-to-interference-plus noise ratios (SINR) for allowing the establishment of reliable communication links. Thus, this research paper starts by reviewing the literature on in-band full-duplex systems for 5G with apache spark and basic concepts of multiple-input multiple-output (MIMO) wireless 5G communications, which allows it to study and propose new techniques and systems where the performance of full-duplex transmissions can be enhanced. In first place, an in-band full-duplex relay station was formally and fully described, special focus was given to the system impairments that cause residual self-interference with apache spark. In order to mitigate this effect, spatial suppression filters were studied, that take advantage of the extra degrees of freedom brought by MIMO transmissions. The CDF was assessed at the relay and at the final destinations, showing a gain in interference resilience when using several antennas and revealing an optimal relay transmit power. Consequently, an algorithm to find the optimal power allocation scheme that minimizes the system energy efficiency (EE) under a minimum link quality was derived.

Table 1. The comparison with other techniques.

ARTICLE	TECHNIQUE	SUPPORT TO NETWORK
[21]	Apache Hadoop	Network traffic analysis in 5G network
[22]	Apache Kafka	Big Data Streaming Process in 5G
Proposed	Apache Spark	Wireless communication with high network bandwidth

VI. Conclusion

The fifth-generation (5G) network is the latest generation of cellular mobile communications. Performance-wise, it is designed to achieve high data rate, reduced latency, high adjustability, energy savings, high capacity, and support for industries via massive device connectivity often called Internet of Things (IoT). These increased performance and novel technologies introduced by the 5G standard offer new possibilities for multiple industries and applications such as factories and wireless communication. However, harnessing premature technologies without extensive research is exposing devices for external attacks and risking lives in critical applications such as automotive and transportation. Thus it is crucial to investigate the technologies used in the later phases of the development. Apart from the increased speed and low communication latency, one significant advantage in the protocol is the possibility of network slicing. The 5G networks are to be built in a way enabling logical network slices, which can be modified to allow telecom operators to provide a network on an as-a-service basis. This Master's research paper examines the new possibilities 5G networks offer for wireless network

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a

communication with high bandwidth of 10 GHz. The prediction of 5G network bandwidth maximum recorded at 10 GHz on 5 nodes of apache spark at 80% of data trained with total 12 nodes. The main question to answer is will the 5G technology offers sufficient performance for wireless communication. This research paper also examines what challenges does the 5G platform face and does it offer sufficient privacy and security with big data.

REFERENCES

[1] T. X. Tran, A. Hajisami, P. Pandey and D. Pompili, "Collaborative Mobile Edge Computing in 5G Networks: New Paradigms, Scenarios, and Challenges," in IEEE Communications Magazine, vol. 55, no. 4, pp. 54-61, April 2017, doi: 10.1109/MCOM.2017.1600863.

[2] T. X. Tran, M.-P. Hosseini and D. Pompili, "Mobile edge computing: Recent efforts and five key research directions", IEEE COMSOC MMTC Commun.-Frontiers, vol. 12, no. 4, pp. 29-33, 2017.

[3] Q. Wu, G. Ding, Z. Du, Y. Sun, M. Jo and A. V. Vasilakos, "A Cloud-Based Architecture for the Internet of Spectrum Devices Over Future Wireless Networks," in IEEE Access, vol. 4, pp. 2854-2862, 2016, doi: 10.1109/ACCESS.2016.2576286.

[4] C. Wu, Y. Wang and Z. Yin, "Energy-efficiency opportunistic spectrum allocation in cognitive wireless sensor network", EURASIP J. Wireless Commun. Netw., vol. 2018, no. 1, 2018.

[5] I. Al Barazanchi, H. R. Abdulshaheed, S. A. Shawkat, and S. R. Binti, "Identification key scheme to enhance network performance in wireless body area network," *Period. Eng. Nat. Sci.*, vol. 7, no. 2, pp. 895–906, 2019.

[6] K. Montgomery, R. Candell, Y. Liu and M. Hany, "Wireless user requirements for the factory workcell", NIST Rep. Adv. Manuf. Ser.—300-8, 2019.

[7] H. H. A. and H. R. A. I. Al-Barazanchi, Z. A. Jaaz, "Practical application of IOT and its implications on the existing software," *2020 7th Int. Conf. Electr. Eng. Comput. Sci. Informatics (EECSI), Yogyakarta, Indones.*, no. October, pp. 10–14, 2020, doi: 10.23919/EECSI50503.2020.9251302.

[8] J. Kalliovaara, R. Ekman, J. Paavola, T. Jokela, J. Hallio, J. Auranen, et al., "Designing a Testbed Infrastructure for Experimental Validation and Trialing of 5G Vertical Applications", 12th international conference on cognitive radio oriented wireless networks and communications(CROWNCOM 2017), 2017.

[9] F. Hu, B. Chen and K. Zhu, "Full Spectrum Sharing in Cognitive Radio Networks Toward 5G: A Survey," in IEEE Access, vol. 6, pp. 15754-15776, 2018, doi: 10.1109/ACCESS.2018.2802450.

[10] W. Ejaz and M. Ibnkahla, "Multiband Spectrum Sensing and Resource Allocation for IoT in Cognitive 5G Networks," in IEEE Internet of Things Journal, vol. 5, no. 1, pp. 150-163, Feb. 2018, doi: 10.1109/JIOT.2017.2775959.

[11] Dong, X.; Cheng, L.; Zheng, G.; Wang, T. Network access and spectrum allocation in next-generation multi-heterogeneous networks. Int. J. Distrib. Sens. Netw. 2019, 15, 1550147719866140.

[12] IEEE. 802.11-2016-IEEE Standard for Information Technology–Telecommunications and Information Exchange between Systems Local and Metropolitan Area Networks–Specific Requirements-Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications; IEEE: Piscataway, NJ, USA, 2016.

[13] H. H. Abbas, Z. A. Jaaz, I. Al_Barazanchi, and H. R. Abdulshaheed, "Survey on Enhanced Security Control measures in Cloud Computing systems," *J. Phys. Conf. Ser.*, vol. 1878, no. 1, p. 012004, 2021, doi: 10.1088/1742-6596/1878/1/012004.

[14] S. A. Ashraf, I. Aktas, E. Eriksson, K. W. Helmersson and J. Ansari, "Ultra-reliable and low-latency communication for wireless factory automation: From LTE to 5G," 2016 IEEE 21st International Conference on Emerging Technologies and Factory Automation (ETFA), 2016, pp. 1-8, doi: 10.1109/ETFA.2016.7733543.

[15] P. Kolodzy, "Next generation communications: Kickoff meeting", Proc. DARPA, 2001-Oct.

[16] H. R. Abdulshaheed, I. Al Barazanchi, H. T. Jaya, and D. Tunggal, "Smart Solutions Based-On Cloud Computing and Wireless Sensing," *Int. J. Adv. Sci. Technol.*, vol. 28, no. 8, pp. 526–542, 2019.

[17] J. Li, Y. K. Li, X. Chen, P. P. C. Lee and W. Lou, "A Hybrid Cloud Approach for Secure Authorized Deduplication," in IEEE Transactions on Parallel and Distributed Systems, vol. 26, no. 5, pp. 1206-1216, 1 May 2015, doi: 10.1109/TPDS.2014.2318320.

[18] Z. A. Jaaz, S. S. Oleiwi, S. A. Sahy, and I. Albarazanchi, "Database techniques for resilient network monitoring and inspection," *TELKOMNIKA Telecommun. Comput. Electron. Control*, vol. 18, no. 5, pp. 2412–2420, 2020, doi: 10.12928/TELKOMNIKA.v18i5.14305.

[19] B. a. Coll-Perales, J. Gozalvez and J. L. Maestre, "5G and Beyond: Smart Devices as Part of the Network Fabric," in IEEE Network, vol. 33, no. 4, pp. 170-177, July/August 2019, doi: 10.1109/MNET.2019.1800136.

[20] J. Yang, H. He and Y. Qiao, "Network traffic analysis based on Hadoop," 2014 4th International Conference on Wireless Communications, Vehicular Technology, Information Theory and Aerospace & Electronic Systems (VITAE), 2014, pp. 1-5, doi: 10.1109/VITAE.2014.6934437.

[21] I. Abdulshaheed, H. R., Yaseen, Z. T., Salman, A. M., & Al_Barazanchi, "An Evaluation study of WiMAX and WiFi on Vehicular Ad-Hoc Networks (VANETs)," *IOP Conf. Ser. Mater. Sci. Eng. Pap.*, vol. 3, no. 12, pp. 1–7, 2020, doi: 10.1088/1757-899X/870/1/012122.

[22] B. R. Hiraman, C. Viresh M. and K. Abhijeet C., "A Study of Apache Kafka in Big Data Stream Processing," 2018 International Conference on Information , Communication, Engineering and Technology (ICICET), 2018, pp. 1-3, doi: 10.1109/ICICET.2018.8533771.

rized licensed use limited to: AMRITA VISHWA VIDYAPEETHAM AMRITA SCHOOL OF ENGINEERING. Downloaded on December 04,2025 at 08:40:22 UTC from IEEE Xplore. Restrictions a